

Information Fusion

Cross-Modal Prompt Learning and Dual-Branch Knowledge Transfer for Cloth-Changing Person Re-identification --Manuscript Draft--

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Abstract:	Cloth-Changing Person Re-Identification (CC-ReID) is a highly challenging task in real-world surveillance scenarios, where the performance of traditional methods is severely limited by significant appearance variations caused by clothing changes. To address this issue, we propose a novel framework, PLDK-ReID, based on Cross-Modal Prompt Learning and Dual-Branch Knowledge Transfer. First, we design a Context-Regularized Prompt Learning strategy, which explicitly disentangles identity and clothing information at the language prompt level through a novel "identity-exclusive" and "global-shared" dual-slot mechanism. Second, we construct a semantics-parsing-based dual-branch collaborative architecture that processes global contextual features and clothing-invariant biometric features in parallel to obtain a more robust visual representation. Finally, we introduce a knowledge distillation mechanism, where the stable biometric branch acts as a teacher to guide the global branch in learning to resist appearance noise. Extensive experiments on several mainstream CC-ReID datasets, including PRCC, LTCC, and VC-Clothes, demonstrate that our proposed PLDK-ReID method achieves superior performance, surpassing various existing state-of-the-art methods. The code will be made available at: https://github.com/ykding666/PLDK.

Cover Letter

Dear Editor,

We are pleased to submit our original research manuscript entitled "**Cross-Modal Prompt Learning and Dual-Branch Knowledge Transfer for Cloth-Changing Person Re-identification**" for consideration for publication in *Information Fusion*.

This paper addresses a long-standing and critical challenge: Cloth-Changing Person Re-identification (CC-ReID). To overcome the appearance interference caused by drastic clothing variations, we propose a novel framework named PLKD-ReID. Through the synergy of three core components—context-regularized prompt learning, a collaborative dual-branch architecture, and knowledge distillation—our framework learns identity representations that are highly robust to appearance changes.

We believe this work is an excellent fit for the aims and scope of *Information Fusion*. Its core contribution is the introduction of a **novel, multi-level information fusion strategy**. Our framework not only fuses linguistic semantics with visual information at a **cross-modal level**, but also fuses global context with local biometric features at an **intra-visual branch level**. Furthermore, we achieve a "knowledge-level fusion" between representations of differing quality via **knowledge distillation**. This end-to-end, collaborative fusion approach provides a new paradigm for solving complex recognition problems characterized by large intra-class variance.

Extensive experiments on three mainstream datasets (PRCC, LTCC, and VC-Clothes) demonstrate that our method significantly surpasses various existing state-of-the-art approaches. We are confident that this research not only offers a high-performance solution to the CC-ReID problem but also provides new insights into cross-modal disentanglement and fusion, holding significant academic value and potential for broader applications in intelligent recognition systems.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved this submission.

Thank you for your time and consideration. We look forward to hearing from you soon.

Sincerely,

Xiangzhou Jian

Columbia University

Responses to Editorial Questions

1. Has your study been compared to the state-of-the-art? Please list 5 key references or justify why not.

Yes, our study has been comprehensively compared against the state-of-the-art (SOTA). In the experiments section of our paper, we quantitatively benchmarked our proposed PLKD-ReID model against 16 competing methods, including 5 general-purpose ReID methods and 11 specialized cloth-changing ReID (CC-ReID) methods, on the mainstream PRCC and LTCC benchmark datasets. Five key references we compared against are:

[1] Xiong M, Yang X, Sun Z, et al. RFFR-Net: Robust feature fusion and reconstruction network for clothing-change person re-identification[J]. Information Fusion, 2025, 118: 102885.

[2] Liu F, Kim M, Ren Z, et al. Distilling clip with dual guidance for learning discriminative human body shape representation[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024: 256-266.

[3] Huang Y, Wu Q, Zhang Z, et al. Meta clothing status calibration for long-term person re-identification[J]. IEEE transactions on image processing, 2024, 33: 2334-2346.

[4] Li S, Sun L, Li Q. Clip-reid: exploiting vision-language model for image re-identification without concrete text labels[C]//Proceedings of the AAAI conference on artificial intelligence. 2023, 37(1): 1405-1413.

[5] Ning E, Li W, Fang J, et al. 3D-Guided Multi-Feature semantic enhancement network for person re-ID[J]. Information Fusion, 2025, 117: 102863.

2. Which public datasets are commonly used for your research problem? Which did you use? If none, please explain.

For the problem of Cloth-Changing Person Re-Identification (CC-ReID), commonly used public datasets in the academic community include PRCC, LTCC, VC-Clothes, and Celeb-reID. In our study, we utilized three of these mainstream and highly representative datasets for a thorough evaluation: **PRCC**, **LTCC**, and **VC-Clothes**. This selection ensures that our assessment of the model's performance is comprehensive and convincing across both real-world scenarios (PRCC, LTCC) and controlled synthetic environments (VC-Clothes).

3. What validation measures are standard in your field? Which did you apply? If none, please explain.

The standard validation measures in the field of person re-identification are the Cumulative Matching Characteristic (CMC) curve and mean Average Precision (mAP). CMC evaluates the probability of finding a correct match within the top-k

ranked results, with Rank-1 accuracy being the most prominent metric. mAP provides a more comprehensive measure of the model's overall retrieval and ranking performance. Our work strictly adheres to this standard, employing both Rank-1 accuracy and mAP as our primary evaluation metrics.

4. What is the main claim of your paper and its significance to the Information Fusion community?

The main claim of our paper is that by employing a synergistic framework that integrates cross-modal prompt learning, a dual-branch visual architecture, and knowledge distillation, we can effectively disentangle identity and clothing features to solve the challenging problem of cloth-changing person re-identification.

Its significance to the Information Fusion community lies in our proposal of a novel, multi-level information fusion strategy to tackle a highly challenging problem. Our framework not only fuses semantic knowledge from language prompts with visual information at a cross-modal level (Textual-Visual Fusion), but also fuses global contextual features with mask-guided local biometric features at an intra-visual branch level (Global-Local Fusion). Furthermore, we achieve a "Knowledge-level Fusion" between visual representations of differing quality via knowledge distillation. This end-to-end, collaborative fusion approach provides a new and effective paradigm for tackling challenging visual recognition tasks characterized by large intra-class variance.

5. What evidence supports your claim? Summarize main results and implications.

Our claim is supported by three pillars of strong evidence:

1. **Comprehensive Quantitative Comparison:** On the PRCC, LTCC, and VC-Clothes public datasets, our method significantly outperforms 16 existing SOTA methods on key metrics. For instance, on the PRCC cloth-changing setting, our method achieves a Rank-1 accuracy of 62.9% and an mAP of 59.6%, comprehensively leading all compared methods.
2. **In-depth Ablation Studies:** We systematically demonstrate that each of our proposed components—Context-Regularized Prompt Learning, the Dual-Branch Architecture, and Knowledge Distillation—makes a solid and progressive contribution to the final performance.
3. **Multi-faceted Visualization Analysis:** Visualizations of feature distance distributions and t-SNE embeddings show that our method learns a more cohesive and separable feature space. Qualitative retrieval results also intuitively demonstrate its superiority in challenging cases.

Together, these results imply that our multi-level fusion framework is a highly effective approach for the CC-ReID problem, and its core principles may inspire

solutions for other recognition tasks with similar challenges.

6. What are the most closely related works by others, and how does your paper relate?

The most closely related works can be divided into two categories. The first includes auxiliary-feature-based CC-ReID methods like FSAM. These methods also attempt to use invariant cues like body shape but typically rely on complex, post-hoc feature fusion modules. While our approach also uses semantic masks, we innovatively employ them to guide a parallel visual branch and fuse information via knowledge distillation, resulting in a more cohesive framework.

The second, and more closely related, category includes other CLIP-based CC-ReID methods like CLIP3DReID. CLIP3DReID leverages CLIP's language capabilities and 3D human priors to learn shape features. In contrast, our method focuses on disentanglement at the language prompt level through a context-regularization strategy and on optimization within the 2D visual feature space via knowledge distillation. We thereby propose a novel pathway to adapt CLIP for CC-ReID that does not depend on 3D information.

7. Have parts of this work been published previously (e.g., conference papers)?

No. This manuscript is an original work. No part of it has been published previously, nor is it under consideration for publication elsewhere.

Highlights

- Propose a novel prompt learning strategy to disentangle identity and clothing.
- A dual-branch architecture fuses global context with biometric features.
- Knowledge distillation transfers clothing-invariant knowledge between branches.
- Outperforms state-of-the-art methods on major cloth-changing ReID benchmarks.

Cross-Modal Prompt Learning and Dual-Branch Knowledge Transfer for Cloth-Changing Person Re-identification

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ABSTRACT

Cloth-Changing Person Re-Identification (CC-ReID) is a highly challenging task in real-world surveillance scenarios, where the performance of traditional methods is severely limited by significant appearance variations caused by clothing changes. To address this issue, we propose a novel framework, PLDK-ReID, based on Cross-Modal Prompt Learning and Dual-Branch Knowledge Transfer. First, we design a Context-Regularized Prompt Learning strategy, which explicitly disentangles identity and clothing information at the language prompt level through a novel "identity-exclusive" and "global-shared" dual-slot mechanism. Second, we construct a semantics-parsing-based dual-branch collaborative architecture that processes global contextual features and clothing-invariant biometric features in parallel to obtain a more robust visual representation. Finally, we introduce a knowledge distillation mechanism, where the stable biometric branch acts as a teacher to guide the global branch in learning to resist appearance noise. Extensive experiments on several mainstream CC-ReID datasets, including PRCC, LTCC, and VC-Clothes, demonstrate that our proposed PLDK-ReID method achieves superior performance, surpassing various existing state-of-the-art methods. The code will be made available at: <https://github.com/ykding666/PLDK>.

1. Introduction

With the rapid deployment of urban surveillance networks and intelligent transportation systems, person re-identification (ReID)[36, 30, 26] has emerged as a critical technology in public security, intelligent retrieval, and video analytics. Contemporary deep learning paradigms typically model ReID as a fine-grained image retrieval problem, leveraging convolutional neural networks (CNNs)[29, 2, 8] or vision Transformers[12, 7] to learn discriminative feature representations. These are often optimized via metric learning, contrastive learning, and related strategies to build end-to-end identity matching systems. However, ReID in real-world scenarios faces serious challenges: significant appearance discrepancies arise due to clothing changes, seasonal variations, and varying camera viewpoints. These issues severely hinder the performance of conventional methods, especially in cross-clothing matching tasks, and pose a substantial limitation to the effectiveness and reliability of ReID systems in practical applications.

To address Cloth-Changing Person Re-Identification (CC-ReID)[5, 4], both academia and industry have undertaken extensive research. Existing approaches can be broadly divided into two categories. The first line of work attempts to exploit relatively stable biometric features—such as gait[17], body contour[1], and skeletal keypoints[27]—for auxiliary identity recognition, as illustrated in Fig. 1. While these methods have demonstrated advantages in capturing

invariant identity traits, they often require additional feature extraction modules, resulting in complex systems with high computational overhead. Moreover, their robustness in real-world scenarios remains underexplored and unverified. The second line of research focuses on feature disentanglement, aiming to decouple identity information from clothing variations at the feature level using techniques such as generative adversarial networks (GANs). However, these methods tend to involve complex and unstable training procedures and heavily rely on the quality of generated data. The limitations of these existing methods motivate the pursuit of more efficient and robust solutions.

In recent years, the remarkable success of vision-language pre-trained (VLP) models—such as CLIP—in image classification, retrieval, and cross-modal learning has inspired researchers to introduce them into the ReID domain, leading to the emergence of Prompt-ReID methods. For instance, approaches like CLIP-ReID utilize fixed or handcrafted text prompt templates to leverage CLIP's image and text encoders for transferring linguistic knowledge into visual representations, thereby improving retrieval performance under standard conditions. However, these approaches typically rely on static, task-agnostic prompts, making them insufficiently expressive to capture identity semantics under clothing variations. They also lack mechanisms for regularizing clothing-induced noise.

To address these shortcomings, we propose a Context-Regularized Prompt Learning strategy tailored to the challenges of CC-ReID, built upon the CLIP-ReID framework. On the text prompt side, we design two types of learnable slots: identity-exclusive and globally shared. The identity-exclusive slots learn independent vectors for each identity

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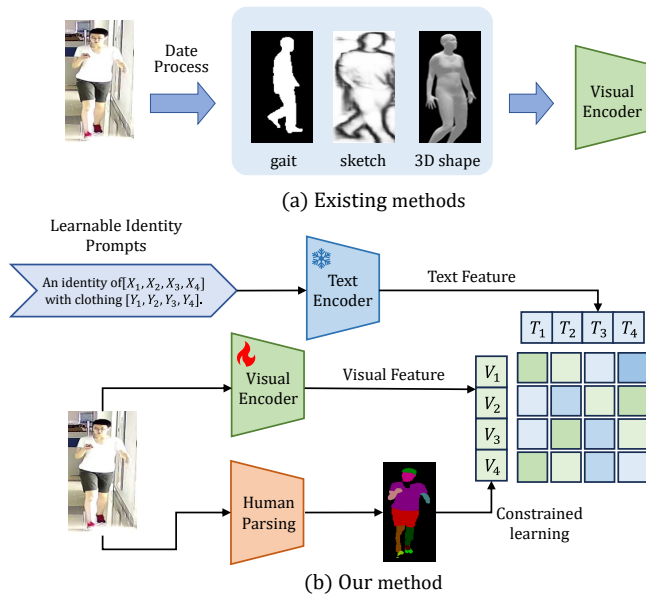


Figure 1: Conceptual comparison between existing CC-ReID methods and our approach. (a) Existing auxiliary-feature-based methods typically rely on additional, separate models to extract clothing-invariant features such as gait, skeletons, or 3D shapes, resulting in complex pipelines. (b) Our PLKD-ReID method performs collaborative learning between learnable text prompts and visual features guided by human parsing within a unified framework.

class to capture pure identity semantics, while the globally shared slots act as strong regularizers. Through shared constraints, they enforce the identity-exclusive slots to disentangle clothing factors, thereby yielding more robust prompt representations. Unlike generic or handcrafted templates, our strategy enables adaptive learning of prompt parameters from data, offering fine-grained modeling for cloth-changing scenarios.

On the visual feature extraction side, we incorporate human semantic parsing[19] tools to generate binary biometric masks, which are processed in parallel with global features from the first $N-1$ layers of the Transformer encoder in a dual-branch architecture. The global branch retains rich contextual information, while the biometric branch focuses on clothing-invariant body parts such as skeletal structure and keypoints, thereby providing more consistent identity cues in cloth-changing conditions. Unlike prior methods that merely concatenate or reweight masked features, our approach enforces strict separation and parallel computation of both branches, which are jointly optimized within a unified framework.

To further enhance the global branch's discriminative capacity in CC-ReID, we treat the biometric branch as the teacher and the global branch as the student, and introduce a distillation-based teacher-student learning mechanism during the second training stage. We adopt temperature-scaled

Kullback-Leibler (KL) divergence to use the soft label distribution of the biometric branch as the teacher signal, guiding the global branch to learn discriminative identity representations free from clothing noise. Through comparative experiments, we demonstrate that this distillation strategy significantly improves cross-clothing matching accuracy and enhances model generalization. The major contributions of this paper are summarized as follows:

- We propose a Context-Regularized Prompt Learning strategy. By introducing a novel dual-slot design—comprising identity-exclusive and globally shared components—we achieve effective disentanglement of identity and clothing factors at the linguistic prompt level, leading to more robust and generalizable prompt representations in cloth-changing scenarios.
- We design a dual-branch collaborative architecture based on semantic parsing, which processes global and biometric features in parallel. Through strict feature decoupling and unified optimization, this framework effectively fuses contextual and clothing-invariant cues, enabling more resilient visual representations.
- We develop a distillation-based knowledge transfer mechanism, where the stable biometric branch acts as a teacher to guide the global branch in learning clothing-invariant identity knowledge. This approach significantly enhances the model's robustness against clothing noise and improves overall performance.
- We achieve state-of-the-art (SOTA) performance on multiple public datasets, including PRCC, LTCC, and VC-Clothes, demonstrating the effectiveness and generalization ability of the proposed PLDK-ReID method.

2. Related Work

2.1. Cloth-Changing Person Re-Identification

Cloth-Changing Person Re-Identification focuses on addressing the drastic appearance variations caused by changes in clothing, which remains a core bottleneck in scaling ReID technologies to real-world applications. Current research efforts in this area generally follow two major paradigms.

The first category leverages stable non-appearance-based biometric cues for auxiliary identification. These methods aim to utilize relatively invariant cues—such as body shape, skeletal structure, or gait—to mitigate the influence of clothing variations. Such body-related features are theoretically clothing-invariant. For instance, FSAM [13] introduces a dual-branch network that processes appearance and body-shape features in parallel and employs attention mechanisms to effectively fuse them, thereby learning representations that are robust to clothing changes. Another representative work, CCPG[25], proposes a GAN-based generative model[21, 6] for data augmentation via clothing and pose transfer. Concurrently, it designs a Relational Graph Attention network to capture cloth-invariant body

shape information, thereby learning more discriminative pedestrian features. However, such methods commonly rely on external pre-trained models, resulting in a cumbersome pipeline. Moreover, in real-world scenarios, the quality of the extracted auxiliary information directly impacts the final performance.

The second line of research concentrates on feature disentanglement. These methods aim to explicitly separate identity-relevant intrinsic features from clothing-related variable attributes in the feature space through carefully designed models. For example, CCFA[10] introduces a novel counterfactual reasoning approach by constructing causal graphs to analyze factor influences and generate counterfactual features, thereby enhancing the model's understanding of essential identity traits. Another notable work, CASE-Net[21], employs generative adversarial networks to swap appearance attributes (e.g., clothing) at the feature level between different identities, synthesizing large volumes of realistic cross-identity samples to improve generalization. Despite their innovative designs, such methods often suffer from training instability and high dependency on the quality and diversity of generated features.

In summary, while existing CC-ReID methods have made significant progress, they still face common limitations. Auxiliary-feature-based approaches heavily rely on the accuracy of external models (e.g., pose estimation, semantic parsing), leading to complex systems and potential error accumulation. Meanwhile, feature-disentanglement and generation-based methods entail intricate and unstable training processes, making it difficult to ensure the authenticity and diversity of generated representations. Therefore, developing a more efficient and robust end-to-end framework that does not rely on complex generative models is key to advancing the CC-ReID field.

2.2. Vision-Language Models in ReID

The emergence of vision-language pre-trained models such as CLIP[28] has introduced a new paradigm for person re-identification, offering powerful semantic alignment capabilities. Researchers have explored various strategies to apply these capabilities to ReID tasks, achieving notable success.

One of the earliest efforts, CLIP-ReID[20], demonstrated the feasibility of directly applying the pre-trained CLIP model to ReID. This approach utilized CLIP's strong image and text encoders to match each person image with a manually designed, learnable textual prompt. Owing to CLIP's superior generalization ability, this simple strategy performed remarkably well across both standard and cross-domain ReID benchmarks, highlighting the potential of VLP models as universal feature extractors.

Subsequent studies extended CLIP's linguistic capabilities to tackle the challenges of CC-ReID. For example, CLIP3DReID [22] presents an advanced method for clothing-changing ReID that harnesses CLIP's language modeling to automatically generate textual descriptors of human body

shapes. The approach aligns the local visual features of a student model with the shape-infused textual features generated by CLIP using optimal transport theory. Simultaneously, it aligns global features with both CLIP's image encoder and a 3D SMPL identity space, thereby learning discriminative body-shape representations that are robust to clothing and viewpoint variations.

These developments reflect an evolution in the application of CLIP to ReID: from serving merely as a feature extractor to being integrated with other modality-specific priors to enhance targeted capabilities. We argue that a more promising direction lies in the deeper adaptation of CLIP's internal mechanisms to precisely address subdomain challenges such as clothing changes. In this work, we follow this trajectory by exploring a novel context-regularized prompt learning strategy and dual-branch knowledge distillation framework, thereby uncovering new potential for VLP models in the CC-ReID domain.

3. Method

To address the significant appearance discrepancy caused by clothing changes in person re-identification (ReID), we propose a novel framework based on context-regularized prompt learning and dual-branch collaborative optimization, as illustrated in Fig. 2. This framework adopts a carefully designed two-stage training strategy, aiming first to learn adaptive textual prompts that can decouple identity and clothing information, and then use these prompts as supervision to deeply optimize a dual-branch visual feature extractor that is robust to clothing variation.

Specifically, the core of the first stage is context-regularized prompt learning. To this end, we design a dynamic template that includes both identity-exclusive and globally shared learnable slots. The key idea is to use the globally shared slots to absorb and generalize the diverse clothing features, thereby constraining the identity-exclusive slots to focus on learning pure and stable identity information, achieving effective regularization of clothing noise.

In the second stage, we freeze the high-quality prompts that have been learned and use them as reliable semantic supervision, shifting the focus to deep optimization of visual features. We construct a semantic-guided dual-branch visual architecture, where the global branch preserves complete contextual information, while the biometric branch leverages semantic masks to focus on body regions that are insensitive to clothing. To further enhance robustness, we introduce a teacher-student distillation mechanism, where the stable biometric branch guides the global branch to learn, transferring clothing-invariant knowledge into the final feature representation, thereby obtaining identity features that are both discriminative and generalizable.

3.1. Context-Regularized Prompt Learning

Conventional CLIP-based ReID methods typically adopt generic and static textual prompt templates such as "a person" or "a photo of a person", which are insufficient to capture fine-grained identity differences in clothing-changing

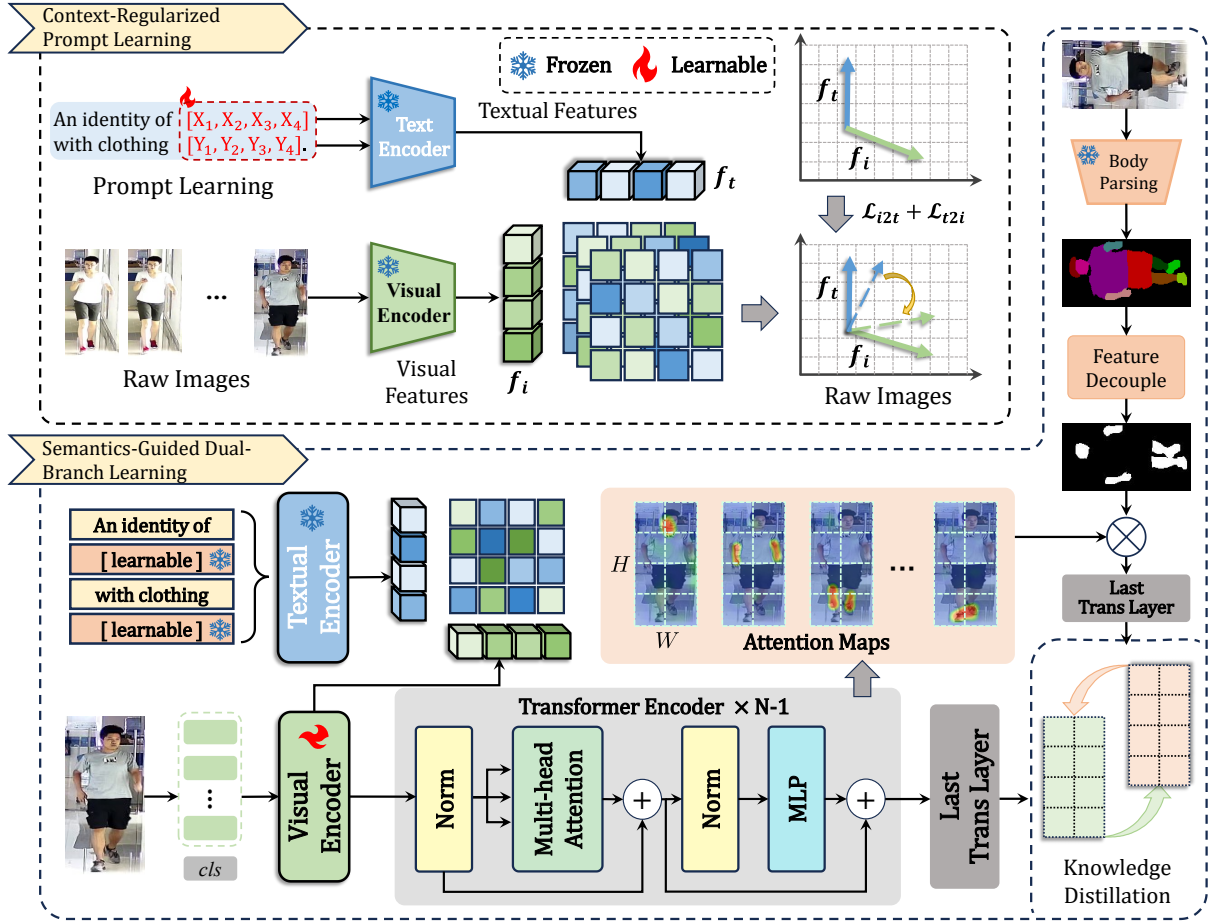


Figure 2: The overall architecture of our proposed PLKD-ReID framework. The framework employs a two-stage training strategy. Stage 1: Context-Regularized Prompt Learning. In this stage, the backbone encoders are frozen to exclusively optimize our designed "identity-clothing" dual-slot prompts for learning disentangled textual representations. Stage 2: Dual-Branch Visual Learning. Here, the text encoder and prompts are frozen to optimize the dual-branch visual encoder. This encoder processes global features and mask-guided biometric features in parallel, and transfers knowledge from the biometric branch to the global branch via distillation.

scenarios. To better exploit the cross-modal matching capability of CLIP pre-trained models and to mitigate feature interference caused by clothing changes, we propose a context-regularized prompt learning strategy, by designing a learnable dual-slot textual prompt template to achieve effective disentanglement and adaptive modeling of identity features and clothing variation information.

Specifically, we design a learnable prompt template that includes identity-exclusive and globally shared slots: "An identity of $[X_1, X_2, X_3, X_4]$ with clothing $[Y_1, Y_2, Y_3, Y_4,]$." Here, "X" denotes the slots that are learned exclusively for each identity class, used to capture personalized, pure, and clothing-invariant identity features; "Y" denotes globally shared slots that are jointly learned across all identity classes, used to capture common information brought by clothing variation. The key idea of this template is to use the globally shared slots to actively absorb and generalize clothing variation patterns, thereby forcing the identity-exclusive slots to focus on intrinsic and clothing-invariant robust identity features.

Formally, for each identity class c , the complete prompt vector P_c can be represented as:

$$P_c = [T_{\text{pre}}; I_c; T_{\text{mid}}; G; T_{\text{suf}}] \in \mathbb{R}^{L \times d} \quad (1)$$

where $T_{\text{pre}}, T_{\text{mid}}, T_{\text{suf}}$ are the fixed prefix, midfix, and suffix embedding vectors; $I_c \in \mathbb{R}^{K \times d}$ is the identity-exclusive learnable prompt slot for identity class c ; and $G \in \mathbb{R}^{M \times d}$ is the globally shared prompt slot. The total prompt sequence length is L , and d is the embedding dimension. During training, we deliberately ignore clothing category labels and adopt a unified globally shared slot. This design forces the model to autonomously learn generalized patterns of clothing variation and effectively eliminate clothing-related interference, thereby enhancing the discriminability of the identity-exclusive slots.

During training, the prompt vector P_c is mapped into the textual feature space via the CLIP text encoder to obtain the textual feature f_c^t , while the input image is mapped into the visual feature space via the CLIP image encoder

to obtain the image feature f_i^v . To fully utilize the cross-modal feature matching capability of the CLIP pre-trained model, we introduce a symmetric bidirectional contrastive loss function, including image-to-text (I2T) and text-to-image (T2I) components:

$$L_{i2t} = -\frac{1}{n_b} \sum_{i=1}^{n_b} \log \frac{\exp(s(f_i^v, f_{y_i}^t)/\tau)}{\sum_{j=1}^{n_b} \exp(s(f_i^v, f_{y_j}^t)/\tau)} \quad (2)$$

$$L_{t2i} = -\frac{1}{n_b} \sum_{i=1}^{n_b} \log \frac{\exp(s(f_i^v, f_{y_i}^t)/\tau)}{\sum_{j=1}^{n_b} \exp(s(f_j^v, f_{y_i}^t)/\tau)} \quad (3)$$

where $s(\cdot, \cdot)$ is the cosine similarity function, τ is a temperature hyperparameter, and n_b is the batch size. The total prompt learning loss is:

$$L_{\text{stage1}} = L_{i2t} + L_{t2i} \quad (4)$$

This symmetric bidirectional alignment optimization mechanism can more effectively drive the fine-grained matching between visual features and textual prompt features in the cross-modal space, ensuring the model learns more robust and clothing-invariant identity representations, thereby improving ReID performance and generalization in clothing-change scenarios.

3.2. Semantic-guided Dual-branch Fusion

To further enhance the robustness of visual features against clothing variations, we propose a semantic-guided dual-branch visual feature extraction architecture. Specifically, for each input pedestrian image, we first use a high-performance human semantic parser to generate a corresponding binary biometric mask that explicitly marks body regions such as the head and limbs which are insensitive to clothing variation. Then, based on a visual Transformer backbone, we build two complementary feature branches in the middle of the network, namely a global feature branch and a biometric feature branch.

In the implementation, given an input image $x \in \mathbb{R}^{3 \times H \times W}$ and its corresponding semantic mask $M \in \{0, 1\}^{1 \times H \times W}$, we first extract intermediate visual feature representations $f_{\text{mid}} \in \mathbb{R}^{(HW+1) \times C}$ through the first 11 Transformer layers of the vision Transformer network. Based on this, we construct the global feature branch and the biometric feature branch.

The global feature branch fully utilizes all spatial feature information of the image by feeding f_{mid} into the 12th Transformer layer to further capture comprehensive contextual information, thereby obtaining global visual features $f_{\text{global}} \in \mathbb{R}^{(HW+1) \times C}$. The features extracted by this branch contain rich scene information but may be significantly disturbed by clothing variations, viewpoint changes, and other factors.

In contrast, the biometric feature branch uses the binary semantic mask M to explicitly constrain the model's attention to visual features in clothing-insensitive regions. Specifically, we perform element-wise multiplication between the

patch features and the mask to explicitly filter out human structural features designated by the mask:

$$f_{\text{bio-input}} = [f_{\text{mid}}^{\text{cls}}; f_{\text{mid}}^{\text{patches}} \odot M] \quad (5)$$

where $f_{\text{mid}}^{\text{cls}}$ denotes the classification token feature, $f_{\text{mid}}^{\text{patches}}$ denotes the spatial features of the image patches, and $\odot$ denotes element-wise multiplication. The filtered features are then fed into a separately designed Transformer layer for further feature extraction, finally producing the biometric visual features $f_{\text{bio}} \in \mathbb{R}^{(HW+1) \times C}$. This branch focuses on human structural regions and therefore provides higher robustness to clothing appearance changes.

Overall, this semantic-guided dual-branch fusion architecture explicitly guides the model to pay more attention to stable feature representations in human body regions, effectively suppressing visual disturbances caused by clothing variation. Compared to solutions that use only global features, our method significantly improves the generalization capability of visual features in clothing-change scenarios and establishes a more robust feature foundation for subsequent teacher-student distillation training and holistic model optimization.

3.3. Teacher-Student Distillation Loss

In the aforementioned dual-branch visual feature extraction architecture, although the global feature branch preserves complete scene semantic information, it is susceptible to interference from visual factors such as clothing variation. In contrast, the biometric feature branch focuses on human structural regions and provides more stable and clothing-invariant identity representations. To fully exploit the robustness of the biometric branch and further enhance the overall visual representation's robustness to clothing variation, we propose a feature distillation strategy based on a teacher-student mechanism. Through explicit knowledge transfer, the global feature branch is effectively guided to learn robust identity cues from the biometric branch.

Specifically, given the visual feature representation f_{bio} from the teacher (biometric) branch and f_{global} from the student (global) branch, we first perform linear projection on the classification token features from both branches to obtain the corresponding class prediction distributions:

$$p_{\text{bio}} = \text{softmax} \left(\frac{W_t f_{\text{bio}}^{\text{cls}}}{T} \right), \quad p_{\text{global}} = \text{softmax} \left(\frac{W_s f_{\text{global}}^{\text{cls}}}{T} \right) \quad (6)$$

where $f_{\text{bio}}^{\text{cls}}$ and $f_{\text{global}}^{\text{cls}}$ are the classification token features of the teacher and student branches respectively, W_t and W_s are their classification weight matrices, and T is a temperature hyperparameter used in the distillation process to soften the output distributions.

To achieve knowledge transfer from the teacher to the student, we define the distillation loss as the Kullback-Leibler (KL) divergence between p_{bio} and p_{global} :

$$L_{KD} = T^2 \cdot KL(p_{bio} \parallel p_{global}) = T^2 \cdot \sum_i p_{bio}^{(i)} \log \frac{p_{bio}^{(i)}}{p_{global}^{(i)}} \quad (7)$$

Under the constraint of this distillation loss, the global branch is explicitly guided to learn clothing-invariant representations encoded in the biometric branch, thereby effectively suppressing disturbances caused by clothing variation and improving the model's generalization capability. The core idea of the distillation process is to utilize the robust identity representation ability of the teacher branch to progressively purify the feature space of the student branch, enabling the model to acquire more pure and robust visual representations.

3.4. Training Loss

To effectively address the challenges of cloth-changing person re-identification, we design a two-stage training strategy. In the first stage, we aim to establish a high-quality association between textual prompts and image content. To achieve this, we freeze the visual and textual encoders and optimize only the embedding vectors of the textual prompts. The training is centered on the bidirectional contrastive loss L_{stage1} :

$$L_{stage1} = L_{i2t} + L_{t2i} \quad (8)$$

where L_{i2t} and L_{t2i} drive the alignment between the visual features and their corresponding textual prompts by maximizing the similarity of matched image-text pairs and minimizing the similarity of mismatched pairs in the shared embedding space.

After completing the alignment in stage one, the training focus shifts in stage two to deep optimization of the visual encoder. We freeze the learned textual prompts and the text encoder, treating them as reliable semantic supervision signals, and introduce a composite loss function to enhance the visual features from multiple perspectives. First, to provide the model with basic identity recognition capability, we introduce an identity classification loss L_{ID} , which applies label-smoothed cross-entropy loss to the global features f_{global} , biometric features f_{bio} , and projection features f_{proj} :

$$L_{ID} = -\frac{1}{N} \sum_{f \in \{f_{global}, f_{bio}, f_{proj}\}} \sum_{i=1}^N y_i \log \hat{y}_i^{(f)} \quad (9)$$

where y_i is the smoothed ground truth label, and $\hat{y}_i^{(f)}$ is the predicted identity probability from each feature branch.

Based on identity classification, to further enhance intra-class compactness and inter-class separability of the features, we introduce a triplet loss L_{Tri} , which applies metric learning constraints to the above three types of features to construct an optimized embedding structure:

$$L_{Tri} = \sum_{f \in \{f_{global}, f_{bio}, f_{proj}\}} \max(0, d(f_a, f_p) - d(f_a, f_n) + \alpha) \quad (10)$$

where f_a, f_p, f_n represent anchor, positive, and negative features respectively, and α is the margin.

Meanwhile, to leverage the alignment established in stage one, we introduce a cross-modal cross-entropy loss L_{i2tce} . This loss treats the textual feature as a ‘‘semantic anchor’’ to supervise the visual features. Specifically, each image's features are projected, and the projected feature f_{proj}^i is matched against the textual features of all identities in the batch, encouraging its highest similarity to the correct identity's text feature $f_{y_i}^t$:

$$L_{i2tce} = -\frac{1}{n_b} \sum_{i=1}^{n_b} \log \frac{\exp(s(f_{proj}^i, f_{y_i}^t)/\tau)}{\sum_{j=1}^{n_b} \exp(s(f_{proj}^i, f_{y_j}^t)/\tau)} \quad (11)$$

where n_b is the batch size, $s(\cdot, \cdot)$ denotes cosine similarity, and τ is the temperature hyperparameter.

Furthermore, in response to the core challenge of the cloth-changing scenario, we propose a novel cross-branch feature consistency loss L_{CFCL} , which aims to guide the model to focus on identity-relevant, clothing-invariant stable features. Specifically, we first stack f_{global} and f_{bio} into a 3D tensor $T = \text{stack}(f_{global}, f_{bio})$, then perform max pooling along the channel dimension to generate a 2D saliency map M that captures the consensus of the two branches:

$$M(i, j) = \max_k T(i, j, k) \quad (12)$$

Then, we promote the consistency of the representations by minimizing the mean squared error between this saliency map M and each branch's features:

$$L_{CFCL} = \frac{1}{2} [\text{MSE}(M, f_{global}) + \text{MSE}(M, f_{bio})] \quad (13)$$

Finally, to further regularize the model and improve generalization, we incorporate the teacher-student distillation loss L_{KD} . This loss leverages the discriminative features learned by a teacher model to guide the training of global features, helping to prevent overfitting during complex multi-task learning and inheriting the generalization capability of the teacher model.

In summary, the overall loss function in stage two is a weighted sum of the aforementioned components:

$$L_{stage2} = L_{i2tce} + \lambda_1 L_{ID} + L_{Tri} + \lambda_2 L_{CFCL} + L_{KD} \quad (14)$$

where λ_1 and λ_2 are hyperparameters to balance the contributions of different losses. This compound loss design

ensures that the model benefits from cross-modal semantic guidance via textual supervision, while also being enhanced by multi-angle self-supervision and internal consistency constraints, ultimately learning highly robust identity representations for cloth-changing person re-identification.

4. Experiment

4.1. Datasets and Evaluation Protocols

We conduct comprehensive evaluations of our proposed PLDK-ReID method on three widely used CC-ReID datasets: PRCC[33], LTCC[27], and VC-Clothes[31]. PRCC is an early benchmark specifically designed for CC-ReID tasks. It contains 33,698 images of 221 identities collected over several months, reflecting real-world clothing variations. LTCC is a more challenging long-term CC dataset, featuring 17,138 images of 152 identities captured from 12 cameras. It presents greater appearance changes over time, including variations in clothing, accessories, and hairstyles. VC-Clothes is a large-scale synthetic dataset generated from the video game Grand Theft Auto V, containing 19,060 images of 512 virtual identities. Its advantage lies in the strictly controlled variables such as clothing, pose, and viewpoint, which facilitate precise evaluation of model performance in clothing-changing scenarios.

To ensure fair performance comparison, we follow the standard ReID evaluation protocols and adopt Cumulative Matching Characteristic (CMC) curves and mean Average Precision (mAP) as the main evaluation metrics. In this paper, we mainly focus on Rank-1 accuracy, which measures the probability that the correct identity appears at the top of the retrieval list. mAP provides a more comprehensive reflection of the model's overall ranking and retrieval performance. According to the official settings of each dataset, our experiments are conducted under three evaluation scenarios: General, where the test set includes all types of images and simulates unconstrained real-world applications; Cloth-Changing (CC), where the test set only includes images with clothing changes to specifically evaluate robustness to appearance variations; and Same-Clothes (SC), where the test set contains only images with the same clothing, serving as a baseline to verify the model's basic identification ability.

4.2. Implementation Details

Our model is implemented using the PyTorch deep learning framework, and all experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU. During training, all input images are resized to 256×128 pixels. To enhance generalization, we apply standard data augmentation techniques including random horizontal flipping, random cropping, and random erasing. The training process is divided into two stages: in the first stage, only the textual prompts are optimized, with a batch size of 128. In the second stage, we perform end-to-end optimization of the visual encoder using an identity-based batch sampling strategy, where each batch randomly selects 16 different identities ($P = 16$), with 4 images per identity ($K = 4$), resulting in a total batch

size of $16 \times 4 = 64$. We use the Adam[18] optimizer with an initial learning rate of 3.5×10^{-4} , and apply a cosine learning rate decay schedule for dynamic adjustment during training. For the loss function, the margin parameter α in the triplet loss is empirically set to 0.3. Other critical loss weighting hyperparameters are discussed in detail in the ablation studies section.

4.3. Comparison with State-of-the-Art Methods

To comprehensively evaluate the performance of our proposed PLDK-ReID model, we compare it with current State-of-the-Art methods on two mainstream CC-ReID datasets: PRCC and LTCC. The comparison includes five general-purpose ReID methods and eleven CC-ReID methods specifically designed for cloth-changing scenarios. Detailed quantitative results are shown in Table 1.

On the PRCC dataset, our method achieves overall superior performance. Under the Cloth-Changing (CC) setting, PLDK-ReID achieves 62.9% Rank-1 accuracy and 59.6% mAP, outperforming all compared methods. It is worth noting that even the generic CLIP-ReID baseline surpasses some early CC-ReID methods, highlighting the potential of CLIP as a strong feature extractor. However, our approach shows more prominent advantages among CLIP-based techniques: compared with CLIP3DReID and SCI, which also build upon CLIP, our model achieves consistently better performance across key metrics, demonstrating the effectiveness of our proposed context-regularized prompt learning and dual-branch knowledge transfer strategies. Furthermore, under the Same-Clothes (SC) setting, our method also reaches top-level performance, indicating that its robustness to clothing variations does not come at the cost of general identification capability.

On the more challenging LTCC dataset, our model also shows strong competitiveness. Under the General setting, it demonstrates excellent generalization ability with 78.9% Rank-1 and 48.9% mAP, outperforming all other methods including MCSC and SCI. Under the Cloth-Changing (CC) setting, our method achieves 50.0% Rank-1 and 26.9% mAP. These results further confirm the success of our CLIP-based framework optimization. Specifically, our Rank-1 performance reaches 50.0%, significantly surpassing the 42.1% achieved by CLIP3DReID and SCI. In terms of mAP, our model scores 26.9%, also clearly exceeding their respective 21.7% and 18.6%. This comprehensive performance improvement further verifies the superiority of our approach.

Additionally, to further evaluate the model's generalization ability, we conduct supplementary experiments on the VC-Clothes dataset, as shown in Table 2. The results indicate that our PLDK-ReID model also achieves leading performance on this dataset, outperforming various existing mainstream methods across key metrics. This confirms the effectiveness and strong adaptability of our method across different datasets and complex clothing-change scenarios.

In conclusion, our PLDK-ReID model achieves highly competitive results across multiple benchmarks, reaching SOTA performance particularly under the CC setting on

Table 1

Comparison With State-of-the-art Methods On PRCC and LTCC.

Method	Reference	PRCC				LTCC			
		SC		CC		General		CC	
		Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
General ReID Method									
PCB[29]	ECCV'18	99.8	97.0	41.8	38.7	65.1	30.6	23.5	10.0
IANet[14]	CVPR'19	99.4	98.3	46.3	45.9	63.7	31.0	25.0	12.6
ISP[37]	ECCV'20	92.8	-	36.6	-	-	-	27.8	11.0
TransReID[12]	ICCV'21	97.3	95.9	47.1	49.3	73.8	39.4	35.7	18.8
CLIP-ReID[20]	AAAI'23	100.0	99.5	57.0	55.3	73.0	36.5	38.3	15.9
CC-ReID Method									
FSAM[13]	CVPR'21	54.5	-	-	-	94.7	94.8	-	-
RCSANet[15]	ICCV'21	50.2	48.6	100	97.2	-	-	-	-
GI-ReID[17]	CVPR'22	33.3	-	-	-	23.7	10.4	63.2	29.4
CAL[9]	CVPR'22	55.2	55.8	100	99.8	74.2	40.8	40.1	18.0
DCR-ReID[3]	TCSVT'23	57.2	57.4	100	99.7	41.1	20.4	76.1	42.3
AIM[34]	CVPR'23	57.9	58.3	100	99.9	40.6	19.1	76.3	41.1
CCFA[10]	CVPR'23	61.2	58.4	99.6	98.7	45.3	22.1	75.8	42.5
CLIP3DReID[22]	CVPR'24	-	-	60.6	59.3	-	-	42.1	21.7
MCSC[16]	TIP'24	99.8	99.8	57.8	57.3	73.9	40.2	42.2	19.4
RFFR-Net[32]	INFFUS'25	100.0	99.9	56.6	57.5	76.0	41.4	41.4	19.8
SCI[11]	Arxiv'25	99.6	97.7	59.8	56.2	75.7	40.6	42.1	18.6
PLKT-ReID(Ours)	-	100.0	99.5	62.9	59.6	78.9	48.9	50.0	26.9

Table 2

Comparison With State-of-the-art Methods On VC-Clothes.

Methods	Venue	VC-Clothes			
		General		CC	
		Rank-1	mAP	Rank-1	mAP
PCB[29]	ECCV18	87.6	74.6	62.0	62.3
ISP[37]	ECCV20	94.5	94.7	72.0	72.1
TransReID[12]	ICCV21	95.1	94.5	70.0	71.8
3DSL[1]	CVPR21	-	-	79.9	81.2
FSAM[13]	CVPR21	-	-	78.6	78.9
GI-ReID[17]	CVPR22	-	-	57.8	64.5
CAL[9]	CVPR22	92.9	87.2	81.4	81.7
AFL[23]	TMM23	93.9	88.3	82.5	83.0
ACID[35]	TIP23	-	-	84.3	74.2
MCSC[16]	TIP24	93.2	87.9	83.3	83.2
PLDK-ReID(Ours)	-	94.8	90.2	86.1	85.9

PRCC and the General setting on LTCC. These excellent results validate the effectiveness of our proposed context-regularized prompt learning, semantic-guided dual-branch architecture, and teacher-student knowledge distillation strategy. The framework successfully disentangles identity and clothing information and learns discriminative representations that are highly robust to appearance changes.

4.4. Ablation Study

To systematically analyze the individual contributions and collaborative effects of the core components in our proposed PLDK-ReID model, we conduct a series of rigorous ablation experiments. To ensure targeted evaluation, all experiments in this section are performed under the most challenging Cloth-Changing (CC) setting on the PRCC and LTCC datasets. We choose the powerful CLIP-ReID as our baseline model and incrementally incorporate the designed modules: Context-Regularized Prompt Learning (CR-PL), Dual-branch Visual Feature Architecture (DBA), and Knowledge Distillation (KD), in a controlled manner. This section aims to thoroughly investigate how each module affects the final performance and to explain the internal mechanisms behind them. Detailed quantitative results are shown in Table 3.

4.4.1. Effectiveness of Context-Regularized Prompt Learning

The effectiveness of the Context-Regularized Prompt Learning (CR-PL) module is preliminarily validated through experiments. By comparing the baseline (Method 1) and the model integrated with CR-PL (Method 2), we observe stable and comprehensive performance improvements. Specifically, on the PRCC dataset, Rank-1/mAP improves from 57.0%/55.3% to 58.5%/56.8%; on the more challenging LTCC dataset, the improvements are even more significant, with Rank-1/mAP increasing from 38.3%/15.9% to

Table 3

Alation State-of-the-art Methods On PRCC and LTCC.

No.	Method	Baseline	CR-PL	DBA	KD	PRCC		LTCC	
						CC		CC	
						Rank-1	mAP	Rank-1	mAP
1	Baseline	✓	×	×	×	57.0	55.3	38.3	15.9
2	Baseline+CR-PL	✓	✓	×	×	58.5	56.8	40.2	18.8
3	Baseline+DBA	✓	×	✓	×	60.2	57.5	45.2	23.9
4	Baseline + CR-PL + DBA	✓	✓	✓	×	61.2	58.1	46.1	24.5
5	Baseline + DBA + KD	✓	×	✓	✓	62.1	58.9	48.4	25.2
6	Baseline + CR-PL + DBA + KD	✓	✓	✓	✓	62.9	59.6	50.0	26.9

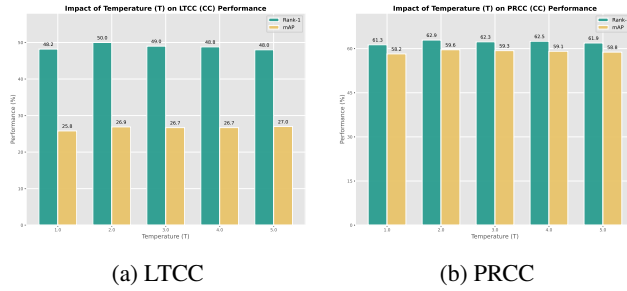


Figure 3: Impact of the temperature hyperparameter (T) in knowledge distillation on model performance.

40.2%/18.8%. This gain stems from our novel dual-slot prompt learning strategy consisting of “identity-exclusive” and “globally shared” slots. The strategy enforces explicit disentanglement between “identity” and “clothing” information within the textual feature space, where the global shared slots absorb and generalize diverse clothing semantics, which in turn “purifies” the identity-exclusive slots. This enables the model to generate more focused and robust identity descriptions, providing high-quality semantic grounding for cross-modal matching and resulting in a foundational improvement in performance.

4.4.2. Influence of Dual-branch Visual Feature Architecture

The Dual-branch Visual Feature Architecture (DBA) is the structural core of the framework, and its introduction brings significant performance gains. Comparing the baseline (Method 1) to the model with DBA alone (Method 3), we observe remarkable improvements—especially on LTCC, where Rank-1 and mAP increase by 6.9% and 8.0% respectively, making it the most impactful standalone module. This substantial performance boost underscores the critical importance of directly and structurally addressing clothing variation at the visual level. DBA constructs parallel global and biometric branches, ensuring that stable body contour features can serve as strong and pure identity cues, unaffected by noisy clothing features. Even more convincingly, this synergy is clearly demonstrated in the comparison between Method 2 and Method 4: by adding DBA on top of

an already optimized prompt, performance is significantly improved again. This indicates that CR-PL and DBA tackle the problem from orthogonal textual and visual dimensions, and that a strong, non-redundant synergy exists between them.

4.4.3. Impact of Knowledge Distillation

Knowledge Distillation (KD) acts as the “refiner” that pushes model performance to the peak and constitutes the final piece of the performance puzzle. Comparing Method 4 to our complete model (Method 6), the inclusion of KD further improves Rank-1 accuracy by 1.7% on PRCC and 3.9% on LTCC, pushing all metrics to their highest values. This clearly shows that even with a powerful dual-branch architecture, the global branch may still retain noise induced by clothing variations. The KD mechanism addresses this by leveraging a teacher–student framework: the more stable and reliable biometric branch acts as the “teacher” to guide the “student” global branch, which is richer in information but more susceptible to noise. Essentially, this process distills the valuable knowledge of clothing invariance from the teacher into the student, effectively suppressing overfitting to clothing details and enabling the student to extract more essential identity features from complete images. This ultimately leads to a significant improvement in generalization.

During the distillation process, the temperature hyperparameter T is critical. As shown in Fig. 3, we conduct a detailed investigation of its effect. A temperature that is too low results in overly “sharp” teacher outputs, resembling hard labels and losing inter-class relational knowledge; conversely, a temperature that is too high produces overly “smooth” distributions, failing to provide effective guidance. Experimental results clearly demonstrate that $T = 2.0$ consistently yields the best performance on both datasets. It achieves a perfect balance between preserving the certainty of the teacher’s knowledge and transmitting inter-class relationships. Therefore, this value is adopted in all our experiments as the final configuration.

4.4.4. Analysis of Hyperparameter Influence

In our composite loss function, the hyperparameters λ_1 and λ_2 are used to balance the contributions of the triplet loss and the cross-branch consistency loss, respectively.

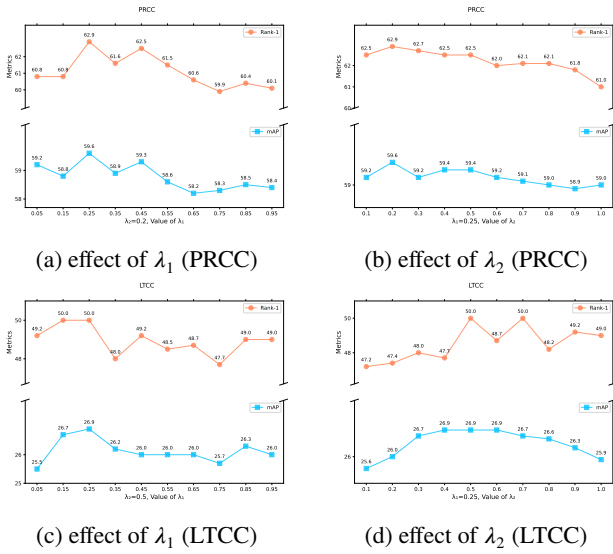


Figure 4: The Impact of Loss Function Hyperparameters λ_1 and λ_2 on the Results of the PRCC and LTCC Dataset.

Their values have a significant impact on the model's final performance. To investigate the model's sensitivity to these two hyperparameters, we conduct a series of experiments on the PRCC and LTCC datasets. We employ a controlled variable method, i.e., when adjusting one hyperparameter, the other is fixed near its optimal value. The experimental results are shown in Fig. 4.

We first examine the effect of the triplet loss weight λ_1 . As shown in the right subplots of Fig. 4, the performance trends on both datasets are highly consistent: as λ_1 increases from 0, both Rank-1 and mAP improve, indicating that the triplet loss, as an effective metric learning constraint, is crucial for constructing a feature space with compact intra-class and well-separated inter-class distributions. However, when λ_1 exceeds a certain threshold (around 0.35), performance begins to plateau or slightly decline, which may be due to overly strong metric learning constraints conflicting with the primary identity classification objective. Overall, $\lambda_1 = 0.25$ achieves optimal or near-optimal performance on both datasets, showing good robustness.

Similarly, we explore the effect of the consistency loss weight λ_2 , as shown in the left subplots of Fig. 4. On the PRCC dataset, performance peaks at $\lambda_2 = 0.2$, then fluctuates and declines as λ_2 increases. On the LTCC dataset, however, the optimal performance is achieved at $\lambda_2 = 0.5$. This phenomenon indicates that due to differences in data distribution and intrinsic difficulty, different datasets have varying sensitivities to the weighting of different loss terms, and thus require targeted fine-tuning. In addition, we validate on the VC-Clothes dataset and find that the combination of $\lambda_1 = 0.55$ and $\lambda_2 = 0.5$ achieves optimal results. This further confirms the effectiveness and generalization capability of our model, while also suggesting that parameter fine-tuning for specific datasets is necessary in practical applications.

4.5. Visualization Analysis

4.5.1. Visualization of Distributions

To more intuitively illustrate how our proposed modules progressively optimize the feature space, we visualize the intra-class and inter-class feature distance distributions of the model at different stages on the more challenging LTCC dataset, as shown in Fig. 5. An ideal ReID model should maximize the inter-class distance and minimize the intra-class distance to enhance feature discriminability. However, as shown in Fig. 5(a), the baseline model is far from ideal in its initial state, with high overlap between the intra-class and inter-class distance distributions, indicating severely insufficient discriminative ability in complex clothing-changing scenarios.

Our proposed modules progressively and effectively improve this situation. As shown in Fig. 5(b), after introducing Context-Regularized Prompt Learning, the means of the two distributions start to separate. Subsequently, integrating the dual-branch architecture brings a qualitative leap: Fig. 5(c) clearly shows the inter-class distances pushed to much higher values, resulting in a significantly enlarged gap between the two distributions. Finally, in our complete PLDK-ReID model, as shown in Fig. 5(d), the mean difference between intra-class and inter-class distributions reaches its maximum, achieving the most effective separation. This progressive optimization process from (a) to (d) visually corroborates the quantitative results in the ablation study, strongly demonstrating that each of our proposed modules contributes solidly and incrementally to improving feature discriminability.

4.5.2. Visualization of t-SNE Embedding

t-SNE[24] (t-distributed Stochastic Neighbor Embedding) visualization is an effective tool for analyzing the structure of high-dimensional feature spaces. Fig. 6 illustrates the feature distributions of several randomly selected identities from the LTCC test set under the baseline model and our proposed PLDK-ReID model. In an ideal feature space, samples from the same identity (denoted by the same color) should form compact, independent clusters, while clusters from different identities should have clear boundaries and sufficiently large separations.

Fig. 6(a) shows the feature distribution learned by the baseline model. The feature clusters are relatively loose, with some identities (e.g., orange and purple) exhibiting highly dispersed intra-class distributions and even a tendency toward splitting. Moreover, the boundaries between different identity clusters are blurred, and the inter-cluster spacing is small, directly explaining the poor performance under clothing-changing scenarios. In contrast, as shown in Fig. 6(b), the feature space learned by our PLDK-ReID model is optimized: each identity forms a highly cohesive and compact cluster, indicating that our model can produce highly consistent representations for different instances of the same identity, even with varying clothing and poses. Meanwhile, inter-identity separability is greatly enhanced, with significantly increased cluster spacing. This visualization strongly

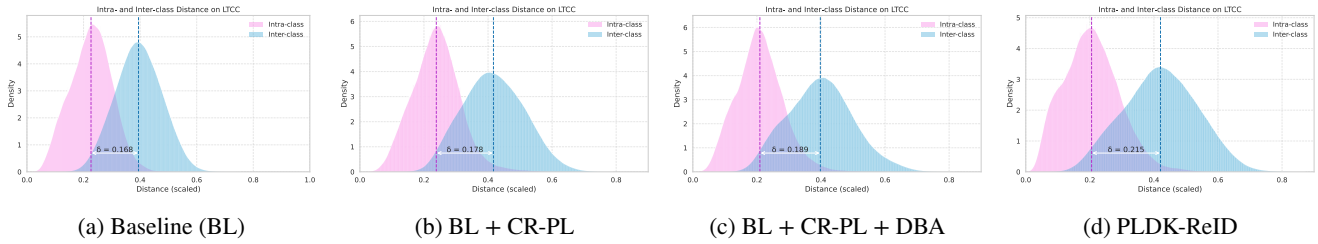


Figure 5: Visualization of intra-class and inter-class distance distributions on the LTCC dataset. The figure illustrates the progressive improvement from (a) the Baseline (BL) to (d) our final model. The pink and blue distributions represent intra-class (same identity) and inter-class (different identities) distances, respectively.

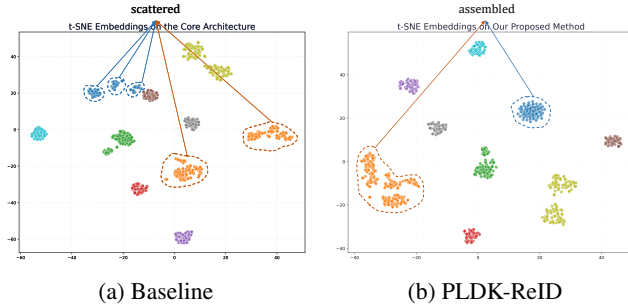


Figure 6: Comparison of t-SNE feature visualizations between the Baseline model and our proposed PLDK-ReID on the LTCC.

demonstrates that our method learns a more discriminative feature space, which is fully consistent with the distance distribution analysis and quantitative experimental results.

4.5.3. Visualization of Retrieval Results

After analyzing the macro-level properties of the feature space through distance distribution and t-SNE, this section presents concrete retrieval cases to demonstrate our model’s practical performance in the final matching task. Unlike the previous analyses of the overall feature space distribution, this visualization focuses on representative and challenging query examples to assess the model’s real-world applicability. Fig. 7 compares the Rank-10 retrieval lists of our PLDK-ReID model and the baseline model on the PRCC and LTCC datasets. In the figure, correct matches—where the retrieved image corresponds to the same person as the query—are marked with green boxes, while incorrect matches are marked with red boxes. It is evident that for each query instance, our method retrieves significantly more correct matches than the baseline within the Rank-10 results, with correct matches generally ranked higher.

On the PRCC dataset, there are substantial clothing changes between query and target images, such as from a dark jacket to a light T-shirt. The baseline model heavily relies on appearance cues like clothing color and texture, leading to many incorrect matches with visually similar but identity-mismatched individuals. In contrast, our PLDK-ReID model successfully learns clothing-invariant features, enabling it to penetrate clothing interference and accurately identify the same person based on more essential biometric

cues such as body shape and contour. On the more challenging LTCC dataset, besides clothing changes, there are severe variations in viewpoint and pose. The baseline model is nearly ineffective under these complex conditions, with retrieval results dominated by incidental and appearance-based mismatches. Our method, however, exhibits remarkable robustness: even when the query image is a side or rear view, it can still accurately retrieve the correct identity from the gallery. In summary, these retrieval visualizations intuitively validate the superiority of our method, aligning closely with the quantitative results reported earlier and strongly supporting the effectiveness of PLDK-ReID in addressing the challenges of cloth-changing person re-identification.

5. Conclusion

In this paper, we address the critical challenge of cloth-changing person re-identification by proposing a novel framework, PLDK-ReID, which integrates cross-modal prompt learning, a dual-branch visual architecture, and knowledge distillation. The framework first employs our proposed context-regularized prompt learning strategy to disentangle identity and clothing information on the language side. Then, leveraging a semantic parsing-based dual-branch architecture, it extracts global contextual features and stable biometric features in parallel on the visual side. Finally, through knowledge distillation, clothing-invariant knowledge is transferred and fused from the biometric branch to the global branch, enabling the learning of highly robust and discriminative feature representations against appearance changes.

Extensive experiments on three public datasets—PRCC, LTCC, and VC-Clothes—including quantitative comparisons with multiple state-of-the-art methods, in-depth ablation studies, and multi-perspective visualization analyses, fully validate the advancement and effectiveness of our proposed method. Although our approach achieves superior performance, future research could evolve from the current implicit disentanglement toward a more fine-grained semantic attribute disentanglement. By decomposing the “globally shared” slots into interpretable semantic slots corresponding to attributes such as “upper-body clothing” and “carried objects,” this design has the potential to enhance the model’s



Figure 7: Qualitative retrieval results comparison between our PLKD-ReID model and the baseline on the PRCC and LTCC datasets. The figure presents several challenging retrieval examples, where green boxes denote correct matches and red boxes denote incorrect ones.

interpretability and enable flexible retrieval based on multi-attribute combinations. This represents a key step toward advancing ReID technology to a higher level of cognitive intelligence.

Data availability

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: